SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

+91-9344249005
rsumanth2005@gmail.com
LinkedIn
GitHub

Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Built a full-stack application using **Java, Spring Boot, MySQL, and React**, developing 5+ REST APIs for business workflows.
  - Designed and optimized **SQL queries, database workflows, and indexing**, reducing backend response time by **25%** during testing.
  - Integrated backend services with frontend components, implementing **API-driven workflows, validation, and database operations** across application modules.

- **Hyphiin** *Apr 2026 – Jun 2026*
  Software Development Engineering Intern Remote
  - Developed **5+ REST APIs** for application workflows, integrating authentication, validation, business logic, and database operations.
  - Contributed to **3+ full-stack modules**, working across backend services, frontend components, and database-driven workflows.
  - Resolved **15+ bugs and integration issues** through debugging and testing, improving application reliability and delivery quality.

Projects

- **Multi-Tenant Financial Services Platform** *2026*
  Technologies: Java, Spring Boot, React.js, TypeScript, PostgreSQL, MongoDB, Docker, AWS, LLMs
  - Built a configurable **multi-tenant application** with role-based access, workflow management, and REST APIs supporting customer and operations use cases.
  - Designed normalized data models and optimized **SQL queries, indexing, and database access** for reliable transaction processing.
  - Integrated an **AI-assisted workflow** for document summarization and information extraction, with validation and exception handling to maintain application correctness.

- **Distributed Job Scheduler** *2026*
  Technologies: Python, FastAPI, PostgreSQL, Redis, Docker, AWS, LLMs
  - Developed a distributed scheduling system with **REST APIs, task queues, worker coordination, retries, and failure handling**.
  - Implemented persistent job tracking, concurrency control, retry mechanisms, and **idempotent execution** using PostgreSQL and Redis.
  - Added an **AI-assisted job monitoring layer** to summarize failures and surface recurring error patterns, with structured logs supporting troubleshooting.

- **AI-Powered SQL Learning Platform** *2025*
  Technologies: React.js, TypeScript, Node.js, SQL, MongoDB, REST APIs, LLMs
  - Built a full-stack platform with **SQL query execution, validation, user progress tracking, and leaderboard** workflows.
  - Developed REST APIs and database operations for **persistent user data, query evaluation, and application state management**.
  - Integrated an **LLM-powered assistant** for contextual SQL hints and query explanations while maintaining deterministic SQL validation and application logic.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Developed and evaluated a deep learning pipeline using a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare.
  - Applied **data preprocessing, PCA, augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ accuracy** and **0.997 AUC**, improving performance by **7% over baseline methods** and MCC by **8%**.

Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

Technical Skills

- **Languages**: Java, Python, C++, JavaScript, TypeScript, SQL
- **Backend Engineering**: Spring Boot, FastAPI, Node.js, REST APIs, API Design, Authentication, Authorization, Microservices
- **Frontend Engineering**: React.js, TypeScript, HTML, CSS, Tailwind CSS
- **Databases**: PostgreSQL, MySQL, MongoDB, NoSQL, Data Modeling, Query Optimization, Indexing
- **Testing / DevOps**: Unit Testing, Integration Testing, Docker, AWS, CI/CD, Git, GitHub, Linux
- **Systems / Core CS**: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design, Distributed Systems, Scalability, Concurrency, Caching
- **AI Engineering**: Generative AI, LLMs, RAG, LangChain, Prompt Engineering, AI-Assisted Development